

MATH 284 Final Review

- 1) Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 2 & 1 & 0 \end{pmatrix}$.
- a) Put A in reduced echelon form by hand.
 - b) If A is the augmented matrix for a linear system, is the system consistent? **Explain.**
 - c) Find all solutions to $A\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. Give your answer in vector form.
- 2) Let A be the matrix from (1). Find the following:
- a) Rank A
 - b) A basis for $\text{Nul } A$
 - c) An orthogonal basis for $\text{Col } A$
 - d) The coordinates of $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ relative to the basis found in (c)
- 3) Find each of the following. Show all steps.
- a) The area of a parallelogram with vertices at $(1, 2)$, $(3, 5)$, $(6, 3)$, and $(8, 6)$
 - b) The matrix of the transformation of $\mathbb{R}^2$ which first rotates clockwise by $\frac{\pi}{6}$ radians, and then expands horizontally by a factor of 4.
 - c) The image under the transformation from (b) of the parallelogram from (a)

- d) The exact area of the transformed image from (c)
- 4) People who buy a domestic car (D) are 60% likely to buy a domestic car the next time, too. The other 40% will buy an imported car (I) the next time. People who buy an imported car are 75% likely to buy an imported car the next time, and 25% likely to buy a domestic car.
- 4) a) If 1000 customers each initially own a domestic car, how many will buy domestic for their 3rd car?
- b) Find the **steady state vector**, and interpret it in the context of this problem.
- 5) Continuing (4), again assume that all 1000 people initially own domestic cars. Find an explicit formula for the exact number who will buy each type for their n^{th} car.
- 6) Evaluate $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 2 & 1 & 2 \end{vmatrix}$ by hand.
- 7) Find the least squares solution to the inconsistent system $\begin{cases} x+2y=8 \\ 2x+y=12 \\ x-y=12 \end{cases}$.
- 8) a) Find the characteristic equation of the matrix $\begin{pmatrix} -5 & 3 \\ -9 & 7 \end{pmatrix}$.
- b) Solve by hand for the eigenvalues and eigenvectors of this matrix.
- c) Diagonalize the matrix.

9) a) Find the general solution to the differential equation $\begin{cases} \frac{dx_1}{dt} = -5x_1 + 3x_2 \\ \frac{dx_2}{dt} = -9x_1 + 7x_2 \end{cases}$

b) Solve the initial value problem if $x(0) = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

c) Carefully sketch the trajectories of the typical solution curves.

10) Find the general solution to the difference equation $y_{k+2} - 5y_{k+1} + 6y_k = 0$

11) Let $T(\mathbf{x})$ be the linear transformation defined by $\mathbf{x} \mapsto A\mathbf{x}$, where $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

a) What are the domain and codomain of T ?

b) Determine whether T is one-to-one, onto, both, or neither.

c) What is the kernel of T ?

12) Find an orthogonal basis for the space W spanned by $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \\ 1 \\ 5 \end{pmatrix} \right\}$ and an orthogonal basis for $W^\perp$.

- 13) What is the distance between the point $(3, 4, 2, -4, 1)$ and the vector space W from the previous problem?
- 14) Note that if $y = Ce^{kx}$ is an exponential model for fitting data, then $\ln(y) = \ln(Ce^{kx}) = \ln(C) + kx$ gives a linear relationship between $\ln(y)$ and x . So, the least-squares line for the data $(x_0, \ln(y_0)), (x_1, \ln(y_1)), \dots$ can be used to find a best-fitting exponential curve for the (x, y) pairs. Use this to find the exponential function $y = Ce^{kx}$ which best fits the data points $(0, 2)$, $(1, 4)$, and $(2, 10)$.
- 15) Orthogonally diagonalize $\begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}$
- 16) Find the maximum value of $-3x_1^2 + 10x_1x_2 - 3x_2^2$ subject to the constraint $x_1^2 + x_2^2 = 1$

- 17) Prove the following is true or give a counterexample to show it may be false: The set of all 2×2 non-invertible matrices forms a subspace of $M_{2 \times 2}$.
- 18) Prove the following is true or give a counterexample to show it may be false: The set of all 2×2 symmetric matrices forms a subspace of $M_{2 \times 2}$.